

Motivation for Spark

- The paradigm of Map-reduce is capable of handling big data by the concept of divide and conquer
- However, in the reduce part, its requirement to collect multiple computational results in a single point takes time
- If an algorithm, e.g., Page Rank, needs to be run iteratively, a map-reduce job is very time consuming
- So Hadoop needs a spark!

In-Memory Processing Speed

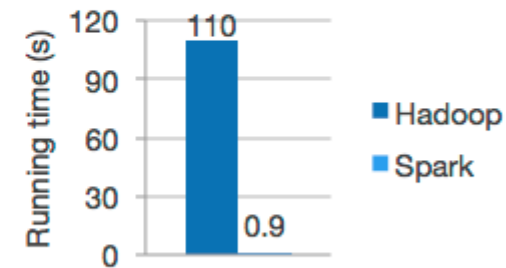
- Access of in-memory processing is 100,000 times faster than disk when performing random access

Apache Spark™ is a unified analytics engine for large-scale data processing.

Speed

Run workloads 100x faster.

Apache Spark achieves high performance for both batch and streaming data, using a state-of-the-art DAG scheduler, a query optimizer, and a physical execution engine.



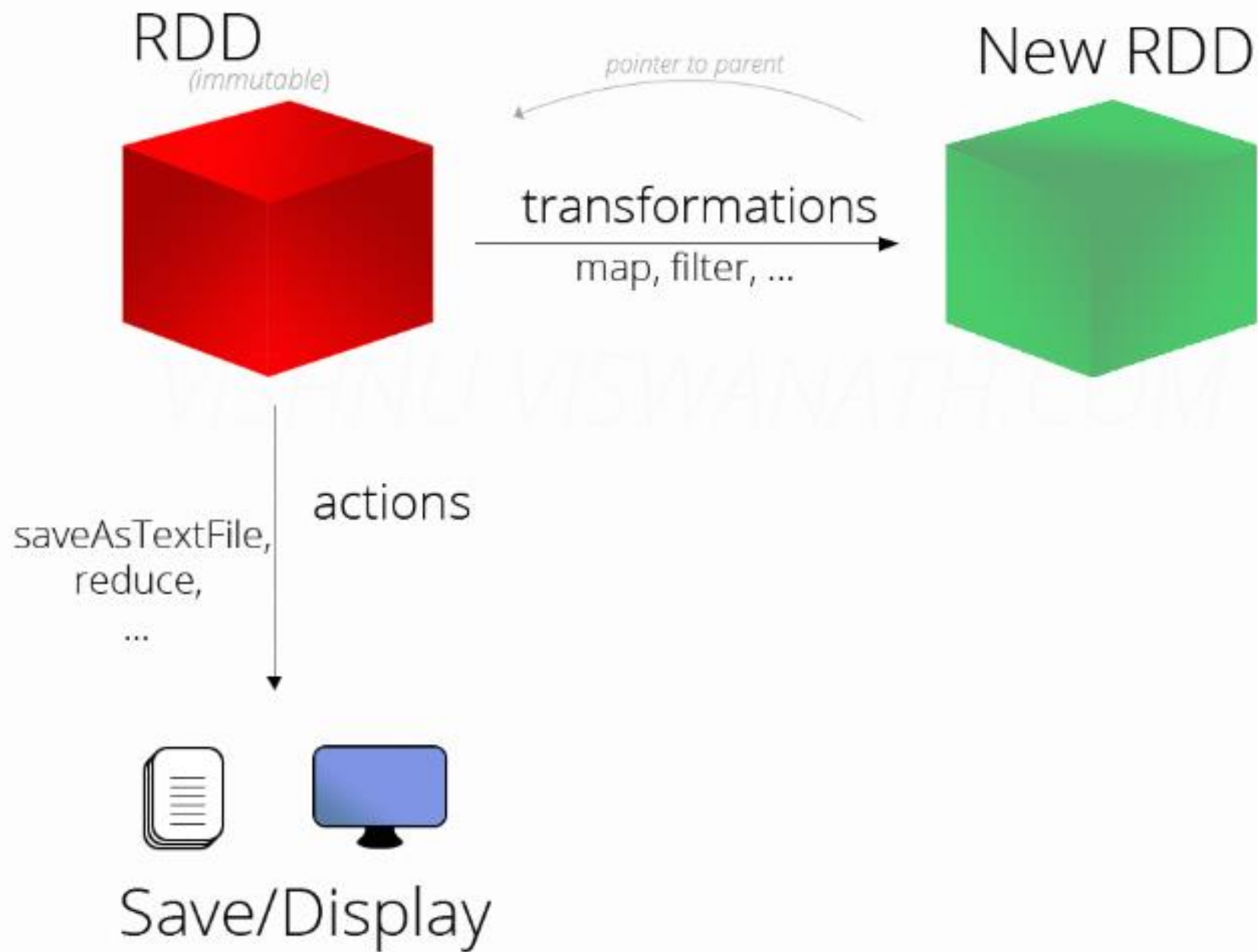
Logistic regression in Hadoop and Spark

What is Spark

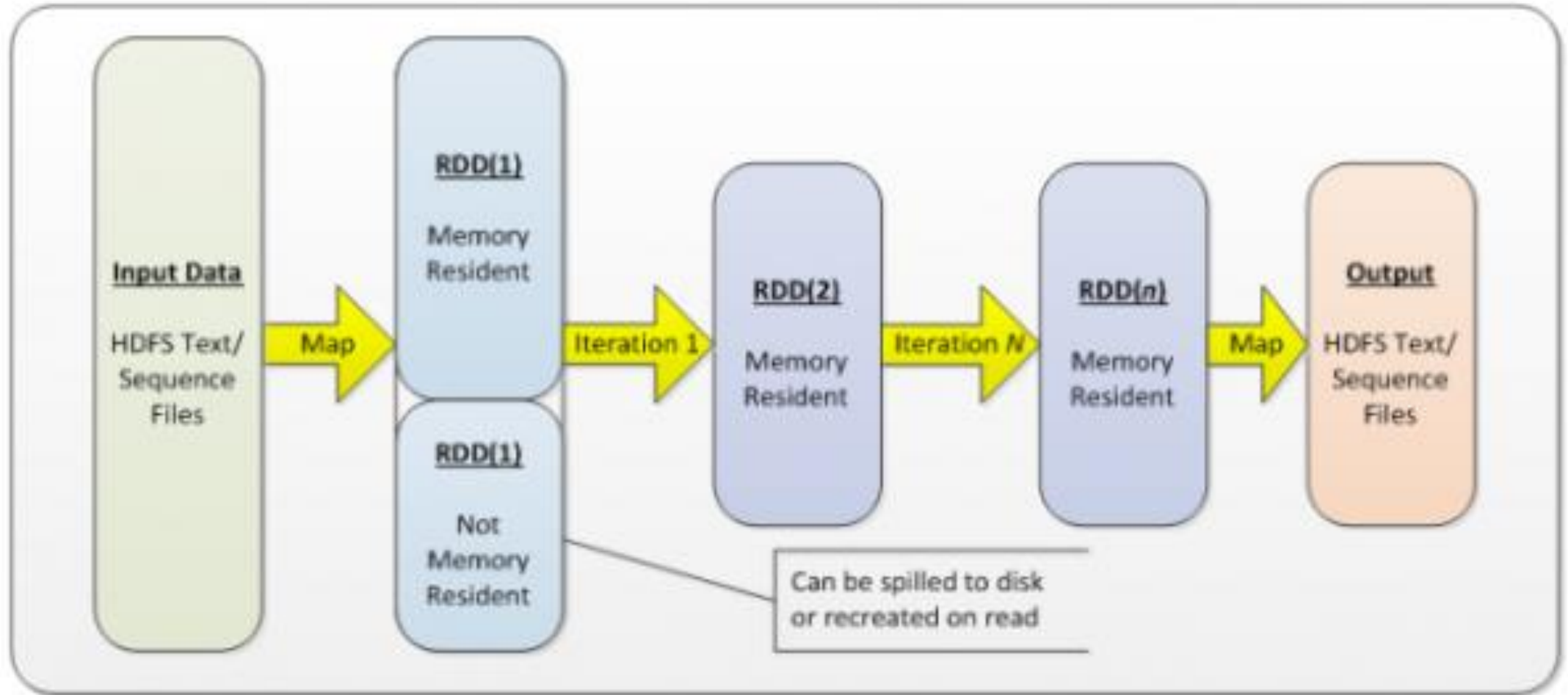
- Developed in UC Berkley AMPLab by Matei Zaharia in 2009
- Spark is written in Scala (Most other Hadoop apps were written in Java)
- An integration of real time and batch computational tasks on HDFS and other data sources
- Spark is an in-memory computational framework so that if a data set can fit into a memory, the computation is much faster (100x)
- Map reduce needs to be done into small pieces in many applications but Spark can be implemented continuously in a sequence of in-memory data
- Spark performs better for iterative algorithms
- <https://www.youtube.com/watch?v=SxAXAhn-BDU>

Inside Spark's Parallel Computation

- RDD (Resilient Distributed Datasets): a programming abstraction that represents a read-only collection of object that are partitioned across a set of machine
- Two types of operations on RDDs:
 - Transformations (like map)
 - Actions (like reduce) e.g. sum or mean
- RDDs are partitioned collections of data that allow the programmer to apply operations to the entire collection in parallel.



Data Management in Spark



A Word Count Example using RDD



Spark: Visualizing RDD transformations

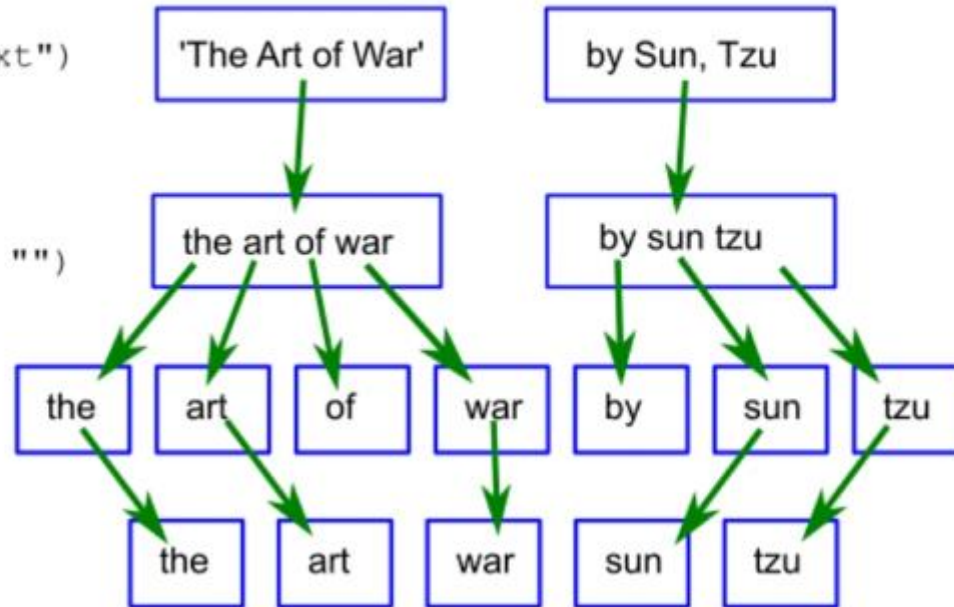
<http://sciabarra.com/blog/spark-visualizing-rdd>

```
sc.textFile("artofwar.txt")
```

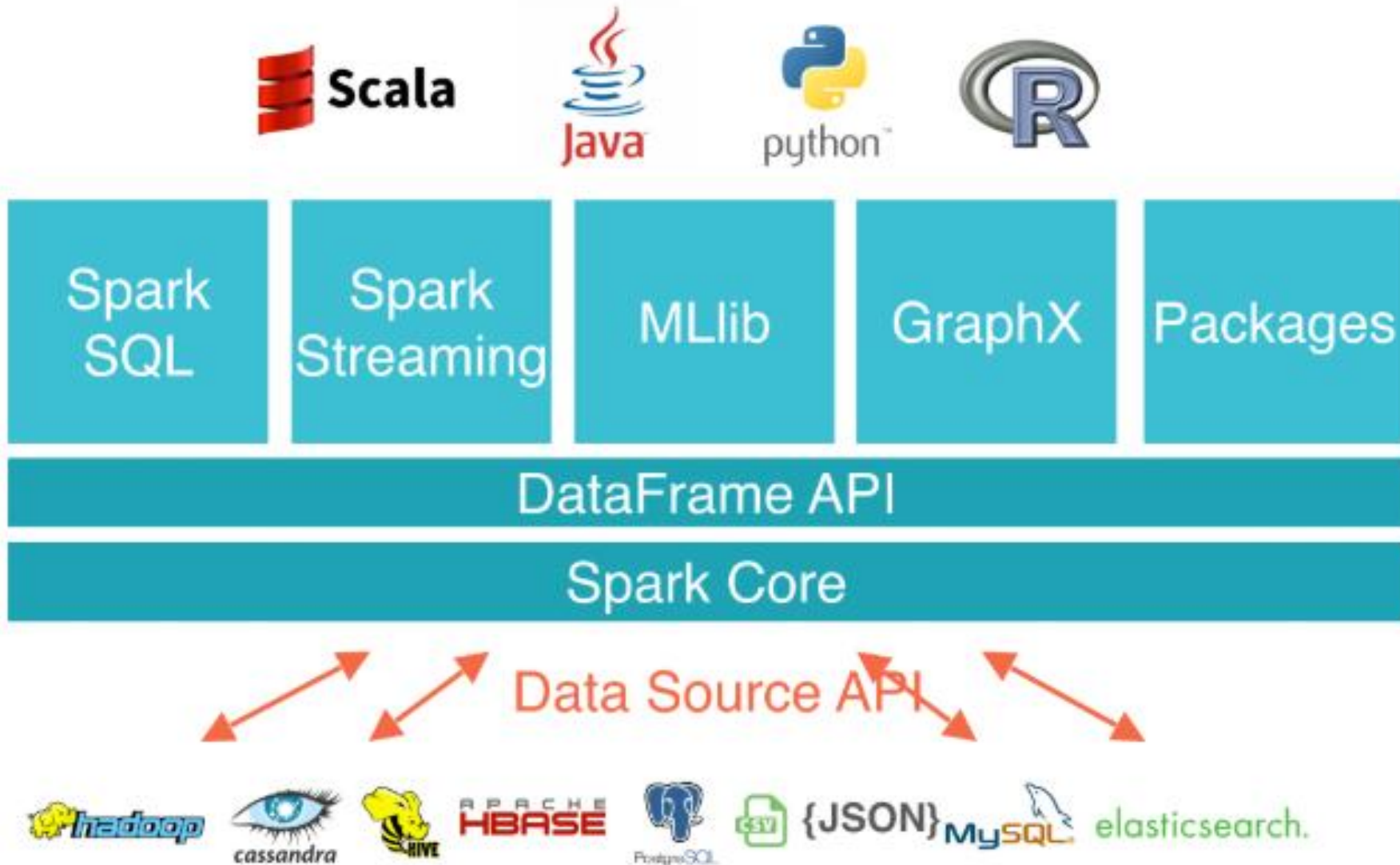
```
.map(  
  _.toLowerCase  
  .replaceAll("[^\\w ]", "")  
)
```

```
.flatMap(_.split(" "))
```

```
.filter(_.size > 2)
```

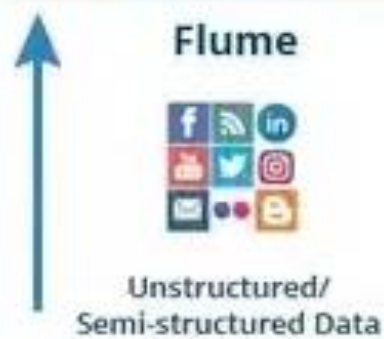


Spark Key Components



Running Spark to Replace Hadoop Ecosystem

- Structured Data Analysis
 - Spark SQL for Hive SQL
- Interactive SQL
 - Spark SQL for Impala
- Machine Learning Analysis
 - MLlib for Mahout
- Real-time Streaming Data Analysis
 - Spark streaming for Storm



Reference

- Chapter 4 of Bengfort and Kim (2016), Data Analytics with Hadoop – An Introduction for Data Scientists, Orilly.
- Spark Tutorial for Beginners
<https://www.youtube.com/watch?v=QaoJNXW6SQo>
- [RDD programming Guide:](#)
 - <https://spark.apache.org/docs/latest/programming-guide.html>